Venkata Lakshmi Parimala Pasupuleti

Data Analyst · Python · SQL · R · Business Intelligence · Data Governance · AWS

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PROFESSIONAL SUMMARY

Data Analyst with 8+ years mining complex, voluminous financial datasets and translating findings into business intelligence tools that drive decisions. Built multi-source regression models achieving **$R^2 = 0.977$** across 190+ countries (Python, R — WHO and UK Road Safety open-source datasets) and classification models reaching **85%+ accuracy on 104K+ records** (Random Forest + Logistic Regression). Designed self-service dashboards for faculty leadership that improved research comprehension **40%** (Tableau, ggplot2, Quarto). At Entrata, tuned **200+ SQL queries for 30% performance improvement** across 1M+ financial records, maintained **100% PHI compliance** with zero violations, and built data quality pipelines with full audit lineage over 7+ years of production operation. Applied Isolation Forest anomaly detection across 3 research pipelines, surfacing outliers manual review missed. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University

Aug 2024 – Aug 2025

Research Associate — Data Analytics · Statistical Modeling · Business Intelligence

- Mined complex, multi-source datasets (WHO global health + UK Road Safety, 190+ countries) using open-source technologies (Python, R) — designed regression models achieving **$R^2 = 0.977$** with end-to-end pipeline: data collection, feature engineering, cross-validation, and automated reporting.
- Surfaced statistically significant anomalies **manual review missed** across 3 research pipelines by applying Isolation Forest — flagged outliers in public health and road safety datasets; translated findings into actionable research recommendations.
- Designed **self-service BI dashboards** (Tableau, ggplot2, Quarto) for faculty leadership — translated complex model outputs into decision-ready visualizations; improved research comprehension **40%** across two cohorts.

Entrata

Jan 2017 – Aug 2024

Software Engineer — Data Analytics · Income Fraud Detection · Financial Data Governance

- Reviewed and verified income documentation for **millions of tenant certifications** across Section 8/HCV, LIHTC, and HUD portfolios — identifying inconsistencies between stated income and supporting documents that flagged ineligible or misrepresented applications for escalation.
- Built **property performance dashboards** across 100K+ unit portfolios — eligibility approval/denial patterns, waitlist demand by bedroom size and income band, occupancy and vacancy rates, rent burden vs. AMI and voucher limits, delinquency rates, and lease renewal trends; applied LIHTC/Section 42 and HUD income-limit rules to validate unit mix, rent limits, and tenant documentation for compliance reporting across Section 8/HCV, HOME, and LIHTC programs.
- Eliminated **80% of manual eligibility review workload** with zero compliance violations by encoding the decision logic I had applied manually into automated Section 8/HCV, LIHTC, and HUD compliance pipelines — AMI threshold calculations, recertification workflows, and escalation routing across 100K+ unit portfolios.
- Achieved **100% PHI compliance** across all client data by migrating to AWS S3 with Libsodium-encrypted pipelines — zero security incidents; improved SQL query performance **30%** across 200+ queries supporting 1M+ property records at 99.5% uptime.

PROJECTS

CrashClarity: Large-Scale Accident Severity Analysis (104K Records)

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Mined 104K+ UK accident records using open-source tools (Python, Scikit-learn) — built Random Forest + Logistic Regression ensemble with feature engineering across weather, road, and temporal variables; achieved **85%+ classification accuracy**; deployed interactive self-service dashboard on GitHub Pages for non-technical stakeholders.

Leasel: Compliance Audit Pipeline with Data Quality Validation

Python · LangGraph · Scikit-learn · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an end-to-end data quality and compliance validation pipeline across 1,240 documents — feature extraction, rule-based classification, and deterministic audit engine achieving **93.6% precision, 87.9% F1**; every determination traceable to its triggering rule with full metadata lineage.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, data governance, ML pipelines, anomaly detection, data visualization

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

Analytics & Modeling: Python, R, SQL, Statistical Modeling, Regression Analysis, Random Forest, Logistic Regression, XGBoost, Isolation Forest, Feature Engineering, Cross-Validation, Hypothesis Testing, Scikit-learn, Pandas, NumPy

Business Intelligence: Tableau, Power BI, ggplot2, Quarto, matplotlib, Jupyter, Interactive Dashboards, Self-Service Data Frameworks, Data Storytelling

Data Governance & Cloud: AWS (S3, EC2, RDS, Glue), Data Quality Management, Metadata & Lineage, PHI/HIPAA Compliance, Audit Pipelines, PostgreSQL, MySQL, PL/SQL, ETL Pipelines, Docker, CI/CD (GitHub Actions), Git, Linux

Process & Additional: Agile/Scrum (120+ sprints), LIHTC/HCV/HUD Regulatory Workflows, Zero-Incident Data Migrations, FastAPI, REST APIs, JavaScript, TypeScript